CAMPUS EVENT FINANCIAL PLANNING

Python Programming Case Study

1. Problem Statement

Organizing a campus event requires careful planning of expenses such as venue rental, equipment, catering, transport, and certificates. Organizers must decide how much participants should contribute, calculate whether sponsorship covers the costs, and determine the minimum registrations required to break even. This program prepares an event budget and supports financial decisions.

2. Objective

To develop a Python program that calculates total event expenses, sponsorship shortfall or surplus, participant contributions, and the minimum registration count needed to cover all event costs.

3. Assumptions

- All costs are entered in the same currency.
- Venue, equipment, and transport are fixed costs.
- Catering and certificates are calculated per participant.
- Sponsorship is a fixed amount.
- Participants share the amount remaining after sponsorship equally.
- Registration count is a positive whole number; costs and sponsorship cannot be negative.

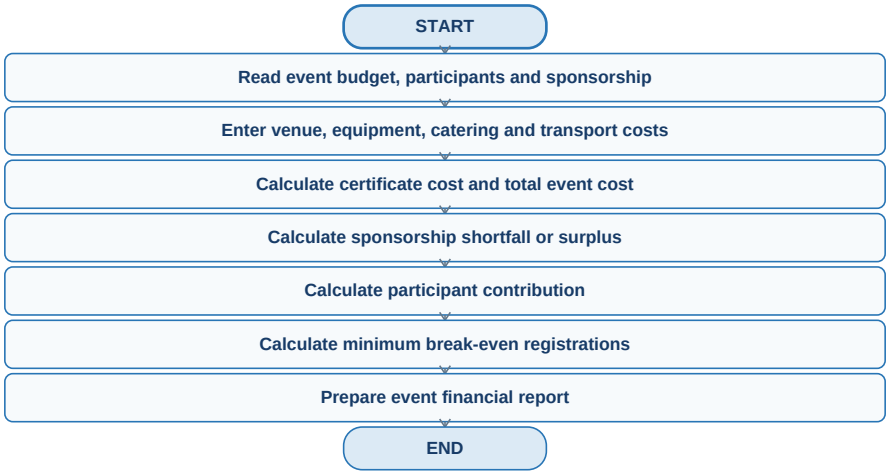
4. Input and Output

Input: Venue, equipment, and transport costs; catering and certificate cost per participant; expected participant count; sponsorship amount; and participant contribution if fixed by organizers.
Output: Fixed cost, variable cost, total event cost, sponsorship shortfall or surplus, contribution per participant, and minimum break-even registration count.

5. Algorithm

1. Start the program.
2. Read venue, equipment, and transport costs.
3. Read catering and certificate cost per participant.
4. Read expected participant count and sponsorship amount.
5. Calculate fixed cost = venue + equipment + transport.
6. Calculate variable cost = (catering + certificate cost per participant) × participants.
7. Calculate total event cost = fixed cost + variable cost.
8. Calculate sponsorship shortfall or surplus.
9. Calculate participant contribution after sponsorship.
10. Calculate the minimum registrations required to cover fixed costs using the contribution remaining after per-person expenses.
11. Display the financial report and stop.

6. Flowchart



7. Pseudocode

```
BEGIN
  INPUT venue_cost, equipment_cost, transport_cost
  INPUT catering_per_person, certificate_per_person
  INPUT participants, sponsorship
  fixed_cost = venue_cost + equipment_cost + transport_cost
  variable_per_person = catering_per_person + certificate_per_person
  variable_cost = variable_per_person * participants
  total_cost = fixed_cost + variable_cost
  balance = sponsorship - total_cost
  IF balance < 0: shortfall = ABS(balance)
  ELSE: surplus = balance
  amount_to_collect = MAX(0, total_cost - sponsorship)
  contribution = amount_to_collect / participants
  net_per_person = contribution - variable_per_person
  IF net_per_person > 0:
    break_even = CEILING(fixed_cost - sponsorship, net_per_person)
  ELSE: break_even = NOT_ACHIEVABLE
  DISPLAY total_cost, shortfall, surplus, contribution, break_even
END
```

8. Calculation Formulae

• Fixed cost = Venue + Equipment + Transport
• Variable cost = (Catering per participant + Certificate per participant) × Participants
• Total event cost = Fixed cost + Variable cost
• Sponsorship balance = Sponsorship – Total event cost
• Participant contribution = MAX(0, Total event cost – Sponsorship) ÷ Participants
• Break-even registrations = Ceiling((Fixed cost – Sponsorship) ÷ (Contribution per participant – Variable cost per participant)), when the denominator is positive.

9. Sample Calculation

Expense	Calculation	Amount
Venue rental	Fixed cost	■10,000
Equipment	Fixed cost	■3,000
Transport	Fixed cost	■2,000
Catering	■150 × 100 participants	■15,000
Certificates	■20 × 100 participants	■2,000
Total event cost	Fixed + variable	■32,000
Sponsorship	Confirmed amount	■12,000
Participant funding	■32,000 – ■12,000	■20,000
Contribution per participant	■20,000 ÷ 100	■200

Break-even example: Fixed costs = ■15,000 and variable cost per participant = ■170. If each participant contributes ■200, ■30 per participant remains toward fixed costs. Minimum registrations = ceiling(■15,000 ÷ ■30) = 500 participants. Thus, 100 registrations are not enough to cover the fixed costs at this contribution level. Organizers must increase the contribution, obtain more sponsorship, or reduce fixed costs. Values are illustrative.

10. Expected Result

The program provides a detailed event budget, calculates the amount to be raised from participants after sponsorship, estimates the contribution per participant, and determines the minimum registration count needed to break even. It helps campus organizers plan expenses, set registration fees, and identify funding gaps.